Experiment 8: Data Visualization I

Title

Data Visualization I

Problem Statement

1. Use the inbuilt dataset **Titanic**.
2. Use Seaborn library to find patterns in the data.
3. Plot a histogram to check distribution of ticket fare (fare column).

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1. Objective

To perform data visualization using:

- Seaborn library
- Titanic dataset
- Histogram plots

and analyze patterns in passenger data.

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2. Theory

What is Data Visualization?

Data visualization means representing data graphically using:

- charts
- graphs
- plots

It helps to:

- identify patterns
- understand trends
- detect outliers
- compare data easily

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3. What is Seaborn?

Seaborn is a Python data visualization library built on top of Matplotlib.

It provides:

- attractive graphs
- statistical plots
- simple syntax

Import statement:

import seaborn as sns

---

4. What is Titanic Dataset?

Titanic dataset contains passenger information such as:

- age
- gender
- class
- fare
- survival status

Total rows:

- 891 passengers

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5. What is Histogram?

A histogram is used to represent distribution of numerical data.

It groups data into intervals called bins.

It helps to:

- understand frequency distribution
- identify skewness
- analyze spread of data

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6. Features of Histogram

- X-axis → data values
- Y-axis → frequency
- Bars touch each other
- Used for continuous data

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7. Procedure

Step 1:

Import required libraries

Step 2:

Load Titanic dataset

Step 3:

Display dataset information

Step 4:

Plot histogram for fare column

Step 5:

Analyze distribution

9. Explanation of Code

Import Libraries

```
import seaborn as sns
import matplotlib.pyplot as plt
```

- seaborn → visualization
- matplotlib → displaying graph

Load Dataset

```
titanic = sns.load_dataset("titanic")
```

Loads Titanic dataset.

Display Dataset

```
print(titanic.head())
```

Displays first 5 rows.

Dataset Information

```
print(titanic.info())
```

Displays:

- column names
- data types
- null values

Plot Histogram

sns.histplot(data=titanic, x="fare", bins=30)

Parameters:

- x = fare column
- bins = number of intervals

Show Graph

plt.show()

Displays histogram.

---

10. Understanding the Histogram

The histogram shows:

- distribution of ticket fare
- frequency of passengers for each fare range

---

11. Observations

Observation 1

Most passengers paid lower ticket fares.

Observation 2

Few passengers paid very high fares.

Observation 3

Data is positively skewed.

Observation 4

Maximum passengers are concentrated in lower fare range.

---

12. Inference

- Most Titanic passengers belonged to lower ticket classes.
- Very few passengers traveled in expensive classes.
- Histogram helps visualize fare distribution clearly.

---

13. Advantages of Histogram

- Easy to understand distribution
- Detects skewness
- Shows frequency
- Helps identify outliers

---

14. Conclusion

We successfully used Seaborn library on the Titanic dataset and plotted histogram for ticket fare distribution to analyze passenger data patterns.

Viva Questions with Answers

1. What is data visualization?

Graphical representation of data.

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2. Why is data visualization important?

It helps understand patterns and trends easily.

---

3. What is Seaborn?

Python visualization library based on Matplotlib.

---

4. What is a histogram?

A graph showing frequency distribution of numerical data.

---

5. What does histogram represent?

Distribution of continuous data.

---

6. What are bins in histogram?

Intervals used to group data values.

---

7. What is Titanic dataset?

Dataset containing passenger information from Titanic ship.

---

8. Which library is used for visualization?

Seaborn.

---

9. Which library is used to display graphs?

Matplotlib.

---

10. What does sns.histplot() do?

Creates histogram plot.

---

11. What is skewness?

Asymmetry in data distribution.

---

12. What is positive skewness?

Data tail extends toward right side.

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13. What does plt.show() do?

Displays graph window.

---

14. What is fare column?

Passenger ticket fare amount.

---

15. Why use histogram?

To analyze frequency distribution.

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16. Difference between bar chart and histogram?

Histogram	Bar Chart
Continuous data	Categorical data
Bars touch	Bars separated

---

17. What is frequency?

Number of occurrences of data values.

---

18. What is continuous data?

Numeric data with range of values.

Example:

- age
- fare
- salary

---

19. What did you learn from this experiment?

How to visualize data using histogram and Seaborn.

---

20. Which dataset was used?

Titanic dataset.

Important Viva Tips

If examiner asks:

“Why are bars touching in histogram?”

Answer:
Because histogram represents continuous data.

“Why use bins?”

Answer:
Bins group data into intervals for better visualization.

“What pattern did you observe?”

Answer:
Most passengers paid low fares and only few paid high fares.

“Why use Seaborn?”

Answer:
Because it provides easy and attractive statistical visualizations.

Experiment 9: Data Visualization II

Title

Data Visualization II

Problem Statement

1. Use the inbuilt dataset **Titanic**.
2. Plot a **box plot** for distribution of **age** with respect to each **gender** along with survival information.
3. Write observations and inference from the graph.

---

1. Objective

To visualize the distribution of passenger ages in the Titanic dataset using a **box plot** and analyze:

- Gender-wise age distribution
- Survival status
- Outliers
- Median and spread of data

---

2. Theory

What is Data Visualization?

Data visualization is the graphical representation of data using:

- charts
- graphs
- plots

It helps:

- understand patterns
- detect outliers
- compare data
- make decisions easily

---

3. What is Seaborn?

Seaborn is a Python visualization library based on Matplotlib.

It is used for:

- statistical plots
- attractive graphs
- easy visualization

Import statement:

import seaborn as sns

---

4. What is Titanic Dataset?

Titanic dataset contains details of passengers such as:

- age
- gender
- ticket fare
- survival status

Important columns:

- sex
- age
- survived

---

5. What is a Box Plot?

A **box plot** is used to display distribution of numerical data.

It shows:

- Minimum value
- Maximum value
- Median
- Quartiles
- Outliers

---

6. Components of Box Plot

Median

Middle value of data.

Quartiles

Divide data into 4 equal parts.

Interquartile Range (IQR)

$$IQR = Q3 - Q1$$

Where:

- Q1 = First Quartile
- Q3 = Third Quartile

---

7. Advantages of Box Plot

- Detects outliers
- Shows spread of data
- Compares distributions
- Easy to understand

---

8. Procedure

Step 1:

Import required libraries

Step 2:

Load Titanic dataset

Step 3:

Plot box plot using Seaborn

Step 4:

Analyze graph

10. Explanation of Code

Import Libraries

```
import seaborn as sns
import matplotlib.pyplot as plt
```

- seaborn → visualization
- matplotlib → displaying graph

Load Dataset

```
titanic = sns.load_dataset("titanic")
```

Loads Titanic dataset from Seaborn.

Create Box Plot

```
sns.boxplot(x="sex", y="age", hue="survived", data=titanic)
```

Parameters:

- x = gender
- y = age
- hue = survival status

Display Graph

```
plt.show()
```

Displays plot window.

---

11. Understanding the Graph

The graph compares:

- Male vs Female passengers
- Survivors vs Non-survivors
- Age distribution

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12. Observations

Observation 1

Female passengers had higher survival rate.

Observation 2

Most passengers were between age 20–40.

Observation 3

Some outliers are present in age distribution.

Observation 4

Children had better survival chances.

Observation 5

Male non-survivors were more compared to females.

---

13. Inference

- Women and children were given higher priority during rescue.
- Survival rate for females was higher.
- Box plot clearly shows variation and outliers in age data.

---

14. Conclusion

We successfully implemented box plot visualization using Seaborn on the Titanic dataset and analyzed age distribution based on gender and survival status.

Viva Questions with Answers

1. What is data visualization?

Graphical representation of data.

---

2. Why is data visualization important?

It helps understand patterns, trends, and outliers.

---

3. What is Seaborn?

A Python visualization library based on Matplotlib.

---

4. What is a box plot?

A statistical graph showing distribution of data.

---

5. What does a box plot show?

- Median
- Quartiles
- Outliers
- Spread of data

---

6. What is median?

Middle value of ordered data.

---

7. What is IQR?

$$IQR = Q3 - Q1$$

Difference between third and first quartile.

---

8. What are outliers?

Values far away from normal data range.

---

9. What is Titanic dataset?

Dataset containing passenger information from Titanic ship.

---

10. Which library is used for plotting?

Seaborn and Matplotlib.

---

11. What does hue parameter do?

Adds extra categorical grouping using colors.

---

12. What is sns.boxplot()?

Function used to create box plot.

---

13. What is matplotlib?

Python library for plotting graphs.

---

14. What does plt.show() do?

Displays graph output.

---

15. Which column represents gender?

sex

---

16. Which column represents age?

age

---

17. Which column represents survival?

survived

---

18. What is an outlier in box plot?

Point outside whiskers.

---

19. Why are box plots useful?

Easy comparison between groups.

---

20. What did you learn from this experiment?

How to visualize and analyze data using box plots in Seaborn.

Important Viva Tips

If examiner asks:

"Why use box plot instead of histogram?"

Answer:
Box plot gives summary statistics and identifies outliers easily.

"Why use hue='survived'?"

Answer:
To compare survivors and non-survivors separately.

"What are whiskers in box plot?"

Answer:
They represent spread/range of data excluding outliers.

"What are quartiles?"

Answer:
Values dividing data into four equal parts.

Experiment 10 — Data Visualization III

Aim

To perform visualization on the Iris dataset using:

- Histogram
- Boxplot
- Outlier detection

and compare feature distributions.

---

1. What is the Iris Dataset?

The Iris dataset is a famous dataset used in Machine Learning and Data Visualization.

It contains:

- 150 flower samples
- 3 flower species:
  - Iris-setosa
  - Iris-versicolor
  - Iris-virginica

Each flower has 4 features:

1. Sepal Length
2. Sepal Width
3. Petal Length
4. Petal Width

Target column:

- Species

---

2. Types of Features

Feature	Type
Sepal Length	Numeric
Sepal Width	Numeric
Petal Length	Numeric
Petal Width	Numeric
Species	Nominal/Categorical

Viva answer:

- Numeric data contains numbers.
- Nominal data contains category names.

---

3. What is Data Visualization?

Data visualization means representing data graphically using charts and plots so patterns, trends, and outliers become easy to understand.

Examples:

- Histogram
- Boxplot
- Scatterplot
- Heatmap

---

4. Histogram

Definition

Histogram is used to show the frequency distribution of numerical data.

It divides data into intervals called bins.

Example:

- How many flowers have petal length between 1–2 cm
- 2–3 cm etc.

Why histogram is used?

- To understand distribution
- To check skewness
- To identify concentration of values

Viva Points

What does histogram show?

It shows frequency distribution of continuous numerical data.

What are bins?

Intervals used to group values.

Which library is used?

Matplotlib / Pandas / Seaborn

Histogram Code Explanation

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv("Iris.csv")

df.hist(figsize=(10,10))
plt.show()
```

Explanation line by line

Code	Meaning
import pandas as pd	Import pandas library
import matplotlib.pyplot as plt	Import plotting library
pd.read_csv()	Load dataset
df.hist()	Create histogram for all numeric columns
figsize	Size of graph
plt.show()	Display graph

Histogram Inference

Sepal Length

- Almost normal distribution

Petal Length

- Clearly separated groups
- Helps classify flower species easily

Petal Width

- Less overlap between species

Important viva statement:

Petal features are more useful for classification than sepal features.

---

5. Boxplot

Definition

Boxplot is a graphical representation using quartiles. It helps detect outliers.

Components of Boxplot

A boxplot contains 5 things:

1. Minimum
2. First Quartile (Q1)
3. Median (Q2)
4. Third Quartile (Q3)
5. Maximum

Important Terms

Quartiles

- Q1 = 25%
- Q2 = 50% (Median)
- Q3 = 75%

IQR

Inter Quartile Range:

$$IQR = Q3 - Q1$$

Why boxplot is used?

- Detect outliers
- Understand spread of data
- Compare distributions

Boxplot Code

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(data=df)
plt.show()
```

Outliers

Definition

Outliers are data points that differ significantly from other values.

Example:
If most petal lengths are between 1–5 and one value is 15, it is an outlier.

Why outliers occur?

- Measurement error
- Data entry mistake
- Natural variation

Methods to Detect Outliers

1. Boxplot

Dots outside whiskers are outliers.

2. Z-score

$$Z = \frac{x - \mu}{\sigma}$$

$$x$$

$$\mu$$

$$\sigma$$

$$z = \frac{x - \mu}{\sigma} \approx 1.2$$

$$\Phi(z) \approx 88.5\%$$

Where:

- x = data point
- μ = mean
- σ = standard deviation

If:

- Z > 3
- Z < -3

then it is considered an outlier.

---

3. IQR Method

Outlier conditions:

$$x < Q1 - 1.5(IQR) \text{or} x > Q3 + 1.5(IQR)$$

Sample Viva Questions with Answers

Q1. Why do we use Iris dataset?

Because it is simple, balanced, and useful for classification and visualization.

Q2. What is histogram?

A graph showing frequency distribution of numerical data.

Q3. What is boxplot?

A plot that shows quartiles, median, spread, and outliers.

Q4. What is an outlier?

A value far away from normal observations.

Q5. What is IQR?

Difference between third quartile and first quartile.

Q6. Which library is used for boxplot?

Seaborn.

Q7. Difference between histogram and boxplot?

Histogram	**Boxplot**
Shows frequency	Shows quartiles
Displays distribution	Detects outliers
Uses bins	Uses median & IQR

Q8. What is median?

Middle value of sorted data.

Q9. What are quartiles?

Values dividing data into four equal parts.

Q10. Which features separate Iris species best?

Petal Length and Petal Width.